design factors was not within the scope of the current study, thus more detailed investigations should be made to pursue the latent opportunities through discrete, tailored composite material structural design.

CONCLUSIONS

The major conclusions of the study are:

1. Attractive V/STOL short haul commercial transports are technically achievable by the 1985 time period.

2. A 100-passenger six fan/six gas generator configuration can be built with a takeoff weight of 100,680 pounds for the 400 nautical mile VTOL mission and 110,020 pounds for the 800 nautical mile STOL mission.

3. A 1.25 FPR system has lower noise and lower direct operating costs for trip distances less than 500 statute miles relative to higher pressure ratio systems.

4. A 97.5 PNdB 500 foot sideline noise level is achievable with projected quiet lift-fan technology.

5. The study goal of 95 PNdB maximum 500 foot sideline noise goal may not be essential to practical introduction of a 1985 V/STOL transport system if V/STOL port arrival and departure areas of the order of 50 to 100 acres can be established that will shield the surrounding population from the high level terminal area noise.

6. Cruise speeds higher than 0.75 mach number do not appear attractive from the direct operating cost viewpoint, however, market capture studies should be accomplished to determine the overall effect of higher cruise speed on short haul airline operations profitability.

7. There are additional technology areas and operational philosophies to be explored that could lead to still more attractive and competitive short haul V/STOL systems.

    a. air cushion landing system
    b. stored energy devices for peak and emergency power
    c. discrete tailored composite structure design
    d. reduced onboard passenger amenities
    e. use of austere intermediate stop facilities.